Highway Engineering

- INS3441 -

PROBLEM SESSION 6



Question – 1

An highway which has 2x1 lanes is going to be designed with a design speed of 90 km/h. The slopes of the red lines of this road section are given as g_1 = 4.5% and g_2 = 2.5% respectively. These red lines intersect at a point with red elevation of 145.0 m and km value of (1+500). Also, due to slope restrictions trucks can only reach 60 km/h speed value.

a) Find the length of the parabolic vertical curve which can be placed on this road section.

b) Take L=350 m and find km values and elevations of the tangents, mid (bisector) point and inflection point.

c) Would the length of the vertical curve change if the road was designed as (2x2) lane?

Given:

$$d_i = 0.2 * V_i + 8$$

$$f = 0.30 \qquad t_r = 1\ s$$

$$S > L \ \rightarrow L = 2 * S - \left(\frac{4.4}{G}\right)$$

$$DN_{km} = T_{1_{km}} + \frac{g_1 * L}{G}$$

$$S < L \ \rightarrow L = \frac{G * S^2}{4.4}$$

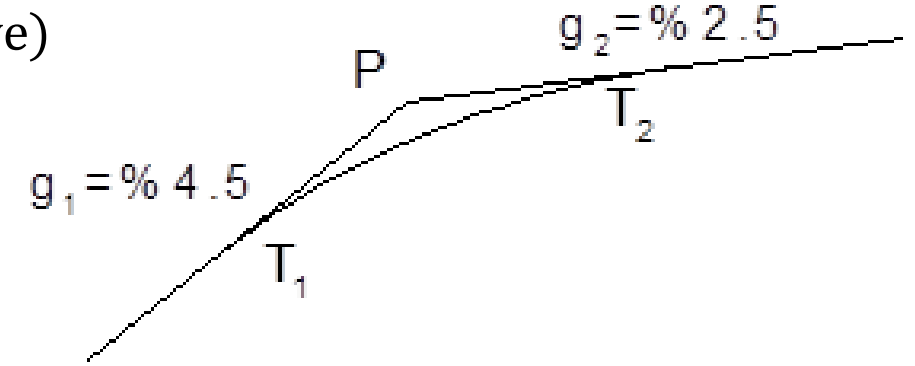
$$DN_{elev} = T_{1_{elev}} + \frac{{g_1}^2 * L}{2 * G}$$

a) Find the length of the parabolic vertical curve which can be placed on this road section.

$g_1 = 4.5\%$
$g_2 = 2.5\%$

$\rightarrow\ G = g_1 - g_2$

$= 0.045 - 0.025$
$= 0.02$ (crest vertical curve)



<u>If the road is not divided</u>; $S = L_s$

$$Ls = \frac{d_1 + d_2}{V_1 - V_2} * (V_1 + V_3)$$

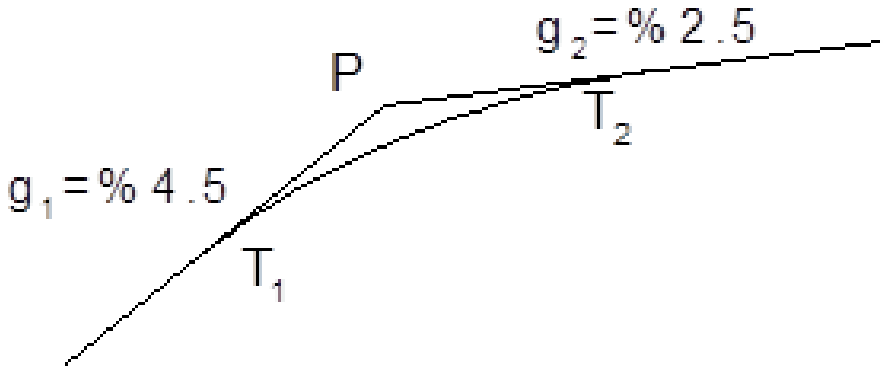
$$Ls = \frac{26+20}{90-60} * (90+90) = 276\ \text{m}$$

$d_i = 0.2 * V_i + 8$

$d_1 = 0.2 * 90 + 8 = 26\ \text{m}$

$d_2 = 0.2 * 60 + 8 = 20\ \text{m}$

$$S = Ls = 276\ m$$



$$S = Ls = 276\ m$$

$$\mathrm{S} > L \ \rightarrow \mathrm{L} = 2 * \mathrm{S} - \left(\frac{4.4}{\mathrm{G}}\right) \ = 2 * 276 - \frac{4.4}{0.02} = 342\ \mathrm{m} \ \rightarrow \ \mathrm{S} = 276 \not> \mathrm{L} = 342$$ ✗

$$\mathrm{S} < L \ \rightarrow \mathrm{L} = \frac{\mathrm{G} * \mathrm{S}^2}{4.4} \ = \frac{0.02 * (276)^2}{4.4} = 346.25\ \mathrm{m} \ \rightarrow \ S = 276 \ < L = 346.25$$ ✓

$$L = 346.25\ m$$

b) Take L=350 m and find km values and elevations of the tangents, mid (bisector) point and inflection point.

$$L = 350\ m \quad \rightarrow \quad e = \frac{G * L}{8} = \frac{0.02 * 350}{8} = 0.875\ m \quad \rightarrow \quad t = \frac{L}{2} = \frac{350}{2} = 175\,m$$

Km Calculation	Elevation Calculation
$T_{1km} = [1 + 500] - 175 = 1 + 325$	$T_{1elev} = P_{elev} - g_1 * t = 145 - 0.045 * 175 = 137.125m$
$T_{2km} = [1 + 500] + 175 = 1 + 675$	$T_{2elev} = P_{elev} + g_2 * t = 145 + 0.025 * 175 = 149.375m$
$B_{km} = P_{km} = 1 + 500$	$B_{elev} = P_{elev} - e = 145 - 0.875 = 144.125\ \ m$
$DN_{km} = T_{1km} + \frac{g_1 * L}{G}$	$DN_{elev} = T_{1elev} + \frac{(g_1)^2 * L}{2 * G}$
$DN_{km} = 1325 + \frac{0.045 * 350}{0.02} = 2 + 112^{50}$	$DN_{elev} = 137.125 + \frac{0.045^2 * 350}{2 * 0.02} = 154.843\ \ m$

c) Would the length of the vertical curve change if the road was designed as (2x2) lane?

If the road is divided; $S = max(l_s ; l_{fe})$

$$l_{fe} = 0.278 * V * t_r + 0.00394 * \frac{V^2}{f \mp s} \quad = 0.278 * 90 * 1 + 0.00394 * \frac{(90)^2}{0.30 - 0.045}$$

$$l_{fe} = 150.17 \ m$$

$$l_s = \frac{d_1 + d_2}{V_1 - V_2} * V_1 \quad = \frac{26 + 20}{90 - 60} * 90 = 138 \ m \qquad S = max(l_s; l_{fe}) = 150.17 \ m$$

$$S > L \ \rightarrow L = 2 * S - \left(\frac{4.4}{G}\right) \quad = 2 * 150.17 - \frac{4.4}{0.02} = 80.34 \ m \quad < S = 150.17 \ m \ \checkmark$$

$$\mathrm{S} < L \ \rightarrow \mathrm{L} = \frac{\mathrm{G} * \mathrm{S}^2}{4.4} \quad = \frac{0.02 * (150.17)^2}{4.4} = 102.50 \ \mathrm{m} \quad \not> S = 150.17 \ m \quad \times$$

Question – 2

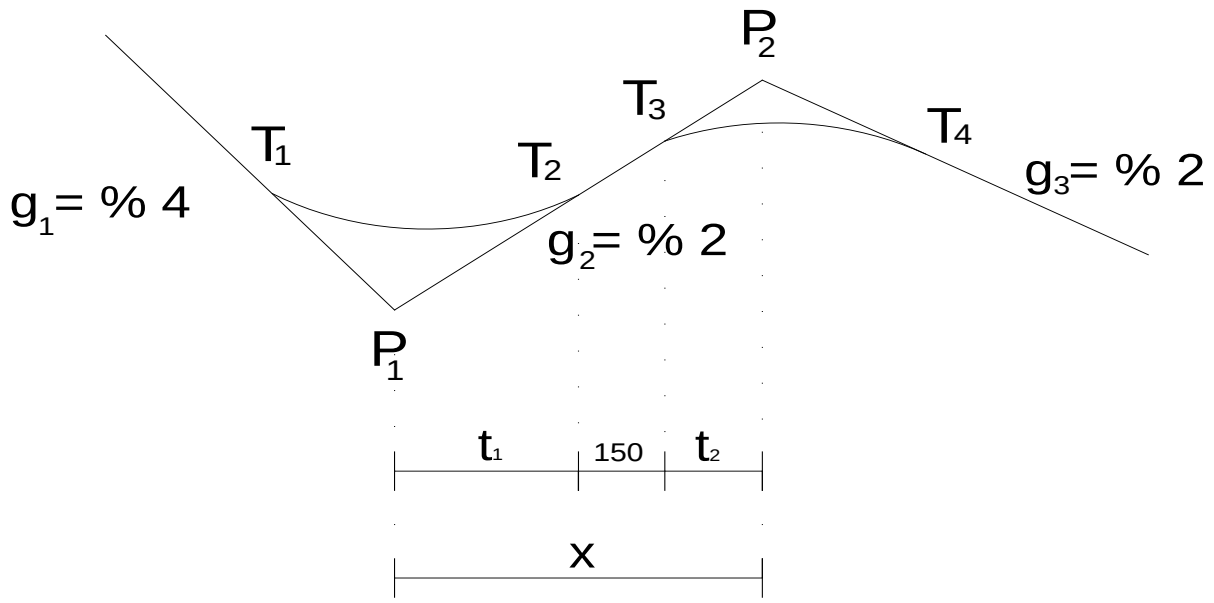
On a road section slopes of red lines are given as $g_1 = -4\%$, $g_2 = 2\%$, $g_3 = -2\%$ and the other values are $P_{1elev} = 350$ m, $P_{1km} = 1+000$, $P_{2elev} = 365$ m. It is required that there should be 150 m between the second tangent point of the first vertical curve and first tangent point of the second vertical curve which will be placed on this road section. Also there is a relation between the tangents of the vertical curves which is $t_1 = 1.5 * t_2$.

a) Calculate the length and radius of both of the vertical curves.

b) Calculate the km values of the mid point, second tangent and inflection point of the first vertical curve.

c) Calculate the red elevations of the first tangent point, mid point and inflection point of the second vertical curve.

a) Calculate the length and radius of both of the vertical curves.



$$0.02 = \frac{P_{2elev} - P_{1elev}}{x} \quad \rightarrow \quad 0.02 = \frac{365 - 350}{x}$$

$$\boxed{x = 750 \ \text{m}}$$

$$t_1 + t_2 + 150 = 750 \ \text{m} \quad \rightarrow \quad t_1 + t_2 = 600 \ \text{m}$$

$$t_1 = 1.5 * t_2 \quad \rightarrow \quad 2.5 * t_2 = 600 \ \text{m}$$

$$\boxed{t_2 = 240 \ \text{m} \ ; \ t_1 = 360 \ \text{m}}$$

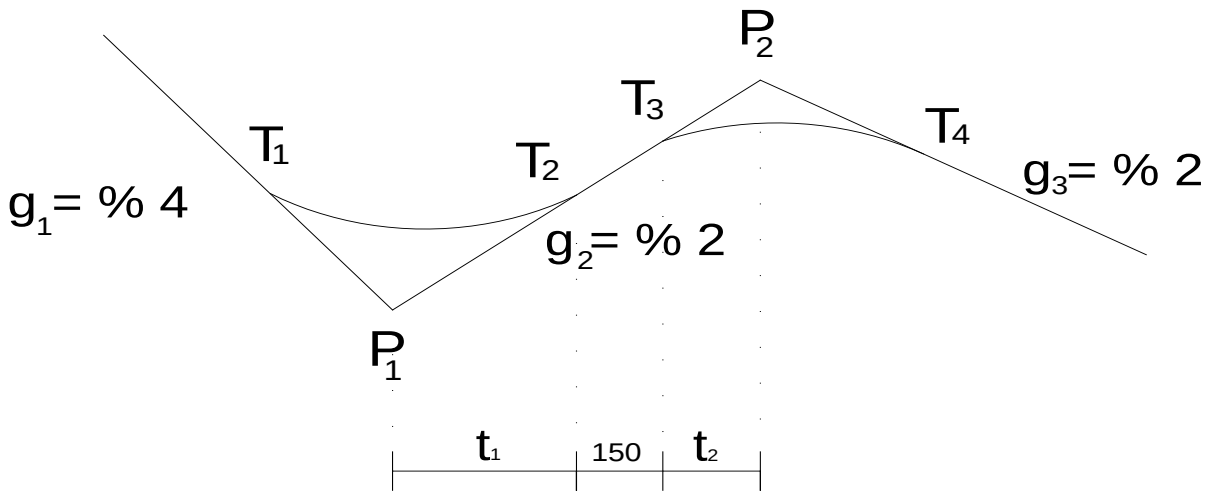
$$L_1 = 2 * t_1 \quad = 2 * 360 = 720 \ \ \text{m}$$

$$G_1 = |-0.04 - 0.02| = 0.06 \quad \rightarrow \ R_1 = \frac{L_1}{G_1} \quad = \frac{720}{0.06} = 12000 \ \ \text{m}$$

$$L_2 = 2 * t_2 \quad = 2 * 240 = 480 \ \ \text{m}$$

$$G_2 = |0.02 - (-0.02)| = 0.04 \quad \rightarrow \ R_2 = \frac{L_2}{G_2} = \frac{480}{0.04} = 12000 \ \ \text{m}$$

b) Calculate the km values of the mid point, second tangent and inflection point of the first vertical curve.

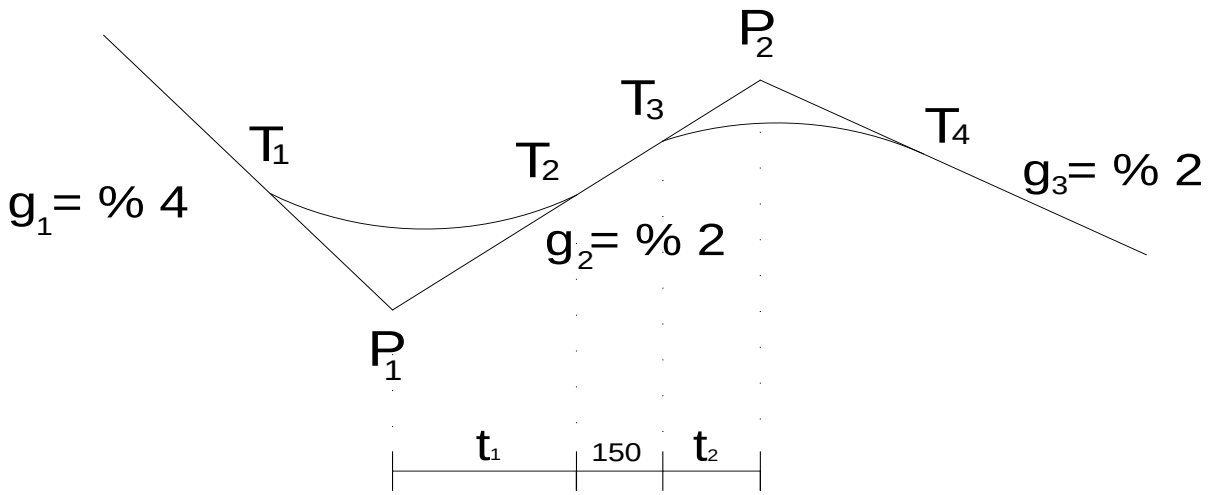


$$B_{1km} = P_{1km} = 1 + 000$$

$$T_{2km} = P_{1km} + t_1 = [1 + 000] + 360 = 1 + 360$$

$$DN_{1km} = T_{1km} + \frac{g_1 * L}{G} = [1 + 000] - 360 + \frac{0.04 * 720}{0.06} = 1 + 120$$

c) Calculate the red elevations of the first tangent pointi mid point and inflection point of the second vertical curve.



$$T_{3elev} = P_{2elev} - g_2 * t = 365 - 0.02 * 240 = 360.20 \ \text{m}$$

$$B_{2elev} = P_{2elev} - e \ = P_{2elev} - \frac{G * L}{8} = 365 - \frac{0.04 * 480}{8} = 362.6 \ \text{m}$$

$$DN_{2elev} = T_{3elev} + \frac{(g_2)^2 * L}{2 * G} \ = 360.2 + \frac{0.02^2 * 480}{2 * 0.04} = 362.6 \ \text{m}$$

Question – 3

On a road section with a design speed of 80 km/h, there are two intersecting vertical alignments at a P point. The slopes are $g_1 = 0.055$ and $g_2 = -0.025$ respectively. Km value of the P point is given as (2+400) m and its elevation is 185 m. It is planned to place a circular vertical curve with a radius of R =5000 m between those vertical alignments.

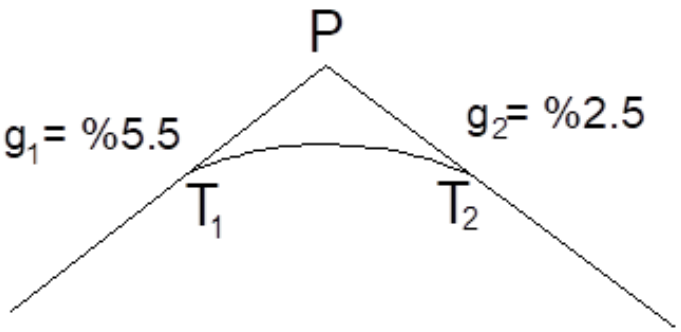
a) Calculate the km values and elevations of the beginning and ending points, mid point and inflection point of the vertical curve.

b) Calculate the red elevations of the cross-sections which are taken at every 50 m after first tangent point of the vertical curve and show them on the table.

Note: Use the table below for calculations.

Cro-Sect. No	Km	x	$r = g_1 * x$	$T_{1elev} \pm r$	$y = K * x^2$	$T_{1elev} \pm r \pm y$

a) Calculate the km values and elevations of the beginning and ending points, mid point and inflection point of the vertical curve.



$$G = g_1 - g_2 = 0.055 - (-0.025) = 0.08 \text{ (Crest Vertical Curve)}$$

$$L = G * R = 0.08 * 5000 = 400 \ m \qquad t = L/2 = 400/2 = 200 \ m$$

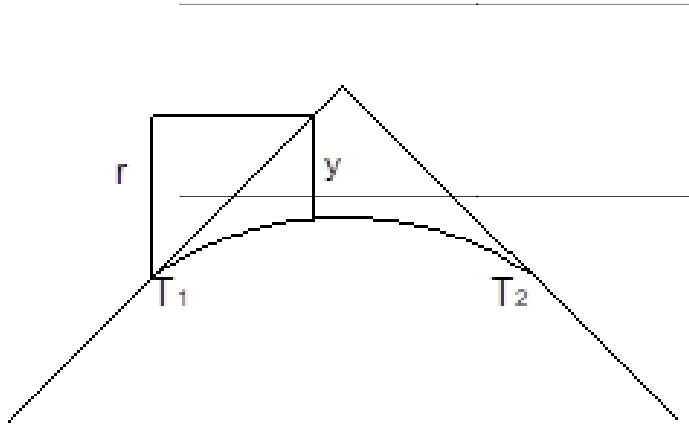
$$e = \frac{G * L}{8} = \frac{0.08 * 400}{8} = 4.0 \ m$$

Km Calculation	Elevation Calculation
$T_{1km} = P_{km} - t = [2 + 400] - 200 = 2 + 200$	$T_{1elev} = P_{elev} - g_1 * t = 185 - 0.055 * 200 = 174$ m
$T_{2km} = P_{km} + t = [2 + 400] + 200 = 2 + 600$	$T_{2elev} = P_{elev} + g_2 * t = 185 - 0.025 * 200 = 180$ m
$B_{km} = P_{km} = 2 + 400$	$B_{elev} = P_{elev} - e = 185 - 4 = 181$ m
$DN_{km} = T_{1km} + \frac{g_1 * L}{G}$ $DN_{km} = [2 + 200] + \frac{0.055 * 400}{0.08} = 2 + 475$	$DN_{elev} = T_{1elev} + \frac{(g_1)^2 * L}{2 * G}$ $DN_{elev} = 174 + \frac{(0.055)^2 * 400}{2 * 0.08} = 181.56$ m

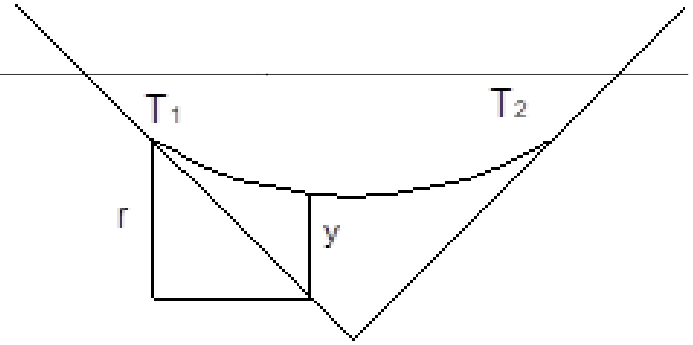
b) Calculate the red elevations of the cross-sections which are taken at every 50 m after first tangent point of the vertical curve and show them on the table.

$$g_1 = 0.055 \qquad y = K * x^2 \qquad K = \frac{G}{2 * L} = \frac{1}{2 * R} = \frac{1}{2 * 5000} = 0.0001 \qquad \rightarrow \qquad y = 0.0001 * x^2$$

Cro-Sect. No	Km	x	$r = g_1 * x$	$T_{1elev} + r$	$y = 0.0001 * x^2$	$T_{1elev} + r - y$



$T_1 + r - y$



$T_1 - r + y$

b) Calculate the red elevations of the cross-sections which are taken at every 50 m after first tangent point of the vertical curve and show them on the table.

$$g_1 = 0.055 \qquad y = K * x^2 \qquad K = \frac{G}{2 * L} = \frac{1}{2 * R} = \frac{1}{2 * 5000} = 0.0001 \qquad \rightarrow \qquad y = 0.0001 * x^2$$

Cro-Sect. No	Km	x	$r = g_1 * x$	$T_{1elev} + r$	$y = 0.0001 * x^2$	$T_{1elev} + r - y$
T_1	2+200	0	0	174	0	174.00
1	2+250	50	2.75	176.75	0.25	176.50
2	2+300	100	5.50	179.50	1.00	178.50
3	2+350	150	8.25	182.25	2.25	180.00
B	2+400	200	11.00	185.00	4.00	181.00
4	2+450	250	13.75	187.75	6.25	181.50
5	2+500	300	16.50	190.50	9.00	181.50
6	2+550	350	19.25	193.25	12.25	181.00
T_2	2+600	400	22.00	196.00	16.00	180.00

Question – 4

On a 2x2 highway with a design speed of 80 km/h there is a parabolic vertical curve which is placed at the intersection of two vertical alignments with slopes g_1 = -3% and g_2 = 5% respectively. Red elevation of the intersection point P is given as 139.5 m and km value of point P is given as (3+200). Find length of the parabolic vertical curve and calculate the elevation and km values of the tangent points and mid point.

Given:

$$d_i = 0.2 * V_i + 6$$

$$f = 0.40 \qquad t_r = 1\ s$$

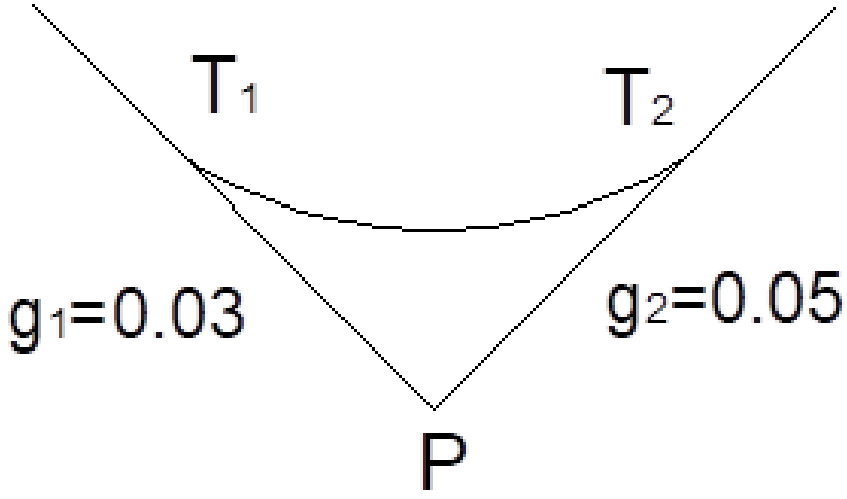
$$S > L \rightarrow L = 2 * S - \left(\frac{1.22 + 0.035 * S}{G}\right)$$

$$V_1 - V_2 = 20\ km/h$$

$$S < L \rightarrow L = \frac{G * S^2}{1.22 + 0.035 * S}$$

Cro-Sect. No	Km	x	r =g_1.x	T_{1kot}±r	y =K.x^2	T_{1kot}±r±y
T_1						
150	3+150					
151	3+175					
B	3+200					
T_2						

$G = -0.03 - 0.05 = -0.08$ (Sag vertical curve)



If the road is divided; $\boldsymbol{S = max(l_s\ ;\ l_{fe})}$

$$l_{fe} = 0.278 * V * t_r + 0.00394 * \frac{V^2}{f \mp s} = 0.278 * 80 * 1 + 0.00394 * \frac{80^2}{0.40 - 0.05}$$

$$l_{fe} = 94.28\ \ m$$

$$l_s = \frac{d_1 + d_2}{V_1 - V_2} * V_1 = \frac{22 + 18}{80 - 60} * 80 = 160 \quad \text{m}$$

$$d_1 = 0.2 * V_1 + 6 = 0.2 * 80 + 6 = 22 \quad \text{m}$$

$$d_2 = 0.2 * V_2 + 6 = 0.2 * 60 + 6 = 18 \quad \text{m}$$

$$S = max(l_s; l_{fe}) = 160\ \ m$$

$$S < L \rightarrow L = \frac{G * S^2}{1.22 + 0.035 * S} = \frac{0.08 * (160)^2}{1.22 + 0.035 * 160} = 300\,m \rightarrow S = 160 < L = 300$$ ✓

$$S > L \rightarrow L = 2 * S - \frac{1.22 + 0.035 * S}{G} = 2 * 160 - \frac{1.22 + 0.035 * 160}{0.08} = 234.75\ m$$ ✘

$$L = 300\ \ m \ \rightarrow \ \ t = \frac{L}{2} = \frac{300}{2} = 150\ m \qquad e = \frac{G * L}{8} = \frac{0.08 * 300}{8} = 3\ \ m$$

Km Calculation	Elevation Calculation
$T_{1km} = P_{km} - t = [3 + 200] - 150 = 3 + 050$	$T_{1elev} = P_{elev} - g_1 * t = 139.5 + 0.03 * 150 = 144m$
$B_{km} = P_{km} = 3 + 200$	$B_{elev} = P_{elev} + e = 139.5 + 3 = 142.5\ m$
$T_{2km} = P_{km} + t = [3 + 200] + 150 = 3 + 350$	$T_{2elev} = P_{elev} + g_2 * t = 139.5 + 0.05 * 150 = 147m$

$$K = \frac{G}{2 * L} = \frac{0.08}{2 * 300} = 0.000133$$

Cro-Sect. No	Km	x	$r = g_1 * x$	$T_{1kot} - r$	$y = 0.000133 * x^2$	$T_{1kot} - r + y$
T_1	3+050	0	0.00	144	0.00	144.00
150	3+150	100	3.00	141	1.33	142.33
151	3+175	125	3.75	140.25	2.08	142.33
B	3+200	150	4.50	139.5	3.00	142.5
T_2	3+350	300	9.00	135	12.00	147.00